Generative AI Project Plan - Group 5

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1 Introduction

Misinformation has become a constant in modern society; people receive misleading information aimed at manipulating their choice and thoughts. The first users of this worthless technique are those that are able to influence public opinion, first of all are politicians that spread them out by quick-truth and fake news to get more votes. Our project aims to help people develop critical thinking by letting them talk with all sides of politics, particularly we focus on the politics of the most influential country in the world, the United States. In this chat, users are able to interact with personas resembling representatives of the two biggest political parties in the States, the Republican Party and the Democratic party, and a third persona representing a more grounded and unpolarized point of view. The representatives exhibit the policy of their respective parties on the topic asked by the user. In this way, the user receives polarized information in a controlled setting with a focus on debate and confrontation, hopefully beneficial for critical thinking.

2 Users

This chat is for everyone, this is its main advantage, it is inclined towards all those who want to get more information about political issues (e.g., understand the policy of a party on climate changes) without relying on any possibly manipulated source.

- **Students** are the most important subjects, since their minds are not shaped yet and they must be protected from extremism. This tool encourages them to develop critical thinking and accurately evaluate their sources. Furthermore, it can be useful for class debates and the improvement of their argumentation skills.
- **Voters** and **Learners** willing to fast understand the policy of both sides on a specific topic to make their consideration during election period find a reliable source in this tool.

3 Data

3.1 General Plan

Our data-gathering process follows a systematic pipeline. First, we will define a set of key political topics, such as healthcare, climate change, and housing, that users are likely to ask about. This narrowing of scope allows us to collect the necessary documents for each party without exceeding our project timeline. For building our database we are thinking about two approaches:

- **1. Approach - Prompt Engineering with JSON Database:** For each topic, we will identify relevant documents (see resources section below) and compile these references in a structured format, such as JSON. Importantly, each persona will draw from distinct sources that reflect their particular ideological perspective. Once the sources are identified, we will extract and organize information for each persona-topic combination into a unified data structure. This first small structured database, in JSON format, will serve as the foundation for our persona system. Each entry will contain:
  - The topic identifier (e.g. healthcare)
  - For each persona perspective:
    - A sample question
    - The persona's perspective on the topic
    - Key supporting arguments
    - Source (e.g. URL)

  In this approach, we will insert the collected information directly into a predefined prompt template. This method provides a useful baseline: it helps us become familiar with prompt-engineering techniques and enables us to build an initial prototype. Moreover, by supplying the LLM with structured information and a defined persona, we give the model guidance on how to interact with users and what "character" it should assume. As a next step we will then additionally employ a more sophisticated RAG Approach.

- **2. Approach - RAG information retrieval with vector database:** To build our RAG system, we will create a vector database from the documents collected for each party and topic. We will split them into smaller, information-rich chunks. Each chunk will be converted into an embedding vector and stored as a separate entry in the vector database, along with metadata. The metadata, such as political party, topic, and source, allows us to distinguish between different parties and retrieve only the relevant information for a given persona. This means we can use a single vector database rather than separate databases for each persona. Furthermore the metadata will also support our fact-checking persona, enabling it to reference the exact source of the retrieved information.

3.2 Data format

The raw data we extract will be semi-structured text mainly sourced from websites that possess inherent structural elements. We will scrape text from specified URLs and clean it by removing URLs, hyperlinks, and useless information. To get the data into a structured format, we may use LLM assistance. Our preprocessing pipeline aims to transform this raw text into a standardized, immediately usable format. The key characteristics of our data structure are as follows:

- **Granularity:** Medium granularity. Each data point (article or text chunk about a certain topic from one political spectrum) represents a complete perspective on a single topic. Importantly, each entry is:
  - Topically focused (one policy area or opinion per entry per political spectrum)
  - Self-contained (comprehensible without additional context or cross-references)
- **Connections:** The final data format uses explicit relationships through topic tags (e.g., healthcare, housing, climate change on one political spectrum), enabling systematic retrieval and organization.
- **Completeness:** The text is already fully developed, professional content. Since we source from official news outlets and formally published policy documents, each entry contains well-articulated positions, coherent arguments, and (possibly) supporting evidence.
- **Context:** Rich metadata accompanies each entry, including author information, publication date, political orientation, and policy area classification. This is already given in the raw text data.
- **Heterogeneity:** Uniform output format despite varied sources. While raw data originates from diverse sources as news pages and policy platforms (each with distinct structural conventions), our scraping and conversion pipeline standardizes all content into a consistent JSON format (1. Approach) or embedding/database entry (2. Approach).

3.3 Time Allocation and Organizational Plan

Data collection, preprocessing, and standardization will require substantial time investment. The workload will be distributed in the following tasks:

- **Data retrieval:** Define the topics we will source information for. Identify key documents per topic and party. Write down the specific resources (URLs) we will source the information (text) from.
- **Generate Transcripts (depending on the time left):** Additionally, if we decide to include transcripts from political speeches, one team member could focus on producing these transcripts, supported by a GenAI model.

- **Build a data pipeline:** Develop a text-extraction workflow that scrapes the specified URLs, cleans the retrieved text, and extracts the key information from each document. Then standardize the outputs into a consistent JSON format across all topics and personas (for 1. Approach).
- **Build a vector database:** Use the predefined resources for each topic and persona, along with the existing text-extraction workflow that already handles text cleaning. Then split the text into smaller chunks, generate embeddings for each chunk, and store them in the vector database together with the relevant metadata.

Some of these tasks can be carried out in parallel, this improves workflow efficiency and ensures a clear division of responsibilities.

3.4 Resources

- Election Programs of Rebublicans and Democrats (PDF): https://shorturl.at/MjY6T
- Republican policies:
  - Heritage Foundation: https://www.heritage.org/
  - American Enterprise Institute: https://www.aei.org/
- Democrats policies:
  - Brooking Institute: https://www.brookings.edu/
  - American progress: https://www.americanprogress.org/
- For fact checking:
  - https://www.allsides.com/topics/facts-and-fact-checking
  - https://toolbox.google.com/factcheck/explorer
  - Wikipedia
  - Web Search Agent
  - PubMed for Health Facts: `https://pubmed.ncbi.nlm.nih.gov/`
  - Free Statistics from `https://www.statista.com/`
  - U.S. Government's Open Data: `https://data.gov/`
- Government official news page: https://www.c-span.org/
- Presidental Speech Transcripts: https://millercenter.org/thepresidency/presidential-speeches
- Congressional speech data such as https://data.stanford.edu/congress_text

4 The Problem

In a time where the pollution of the information system is one of the most oppressive dangers for our societies, our system aims to address the problem of incomplete or misleading perspectives on various topics by providing AI-generated opinions from multiple perspectives (personas). Users can interact with the system to explore how different personas (e.g. Republican, Democrat and fact-checker) might respond to specific topics. Therefore, users are able to enrich their own thought process with potentially previously unknown perspectives or opinions and build resilience against polarization.

5 The Solution

5.1 Concept

Our system performs the following tasks:

- **Understand:** Parse and interpret the user's input (e.g. statement, question, topic) and identify key claims or themes.
- **Retrieve:** Identify the relevant personas and relevant data chunks from the JSON structure or RAG corpus.
- **Generate:** Use the retrieved information to generate an opinion-based response that simulates the persona's believes and worldview.

5.2 Current Personas

- Republican
- Democrat
- journalist / fact-checker

5.3 User Interface

The usage of libraries like Chain-lit[1] would provide a complete and complex chatbot interface without requiring development of a front-end back-end architecture that could take time from the main objective of this project: working on generative AI. Would that not be preferable, we could implement a web-based environment that will provide access to our system. Users can enter a topic or question that interests them and select the desired personas (Republican, Democrat or Fact-checker/Journalist).

[1] `https://github.com/Chainlit/chainlit`

5.4 Technical Approach

To achieve our vision of this system, we have to approach several challenges:

1. **Persona Representation:**
    - How can we design the different personas that accurately resemble their real-world examples?
    - How can we design the different personas while avoiding stereotypical or biased views?
2. **Knowledge Inclusion:**
    - How can we include factually correct knowledge into the system?
    - How can we combine the different persona and this knowledge base to work together as one generating system?

6 Evaluation

Extensive manual evaluation on different aspects like Perspective Accuracy ("Does each persona reflect authentic political positioning?"), Fact-Check Coverage ("Does the fact-checker identify and address verifiable claims"), Factual Accuracy ("Are personas factually correct?" the grounded one has to be), Retrieval Relevance ("Does RAG return contextually useful chunks?"). For some of these metrics, employing LLM as a judge, a tecnhique where a bigger LLM is asked to evaluate, could be of assistance.

The fact-check coverage could be assisted by Google's Fact Checking Tools.

Machine metrics could be similarity between answers of the three personas (measuring semantic similarity) and response time (relevant for user usage).

7 Limitations and Possible Developments

7.1 Data Availability

One limitation that we are going to meet inevitably is the limited amount of data to work on. Gathering textual data from reliable sources to build a sufficiently big corpus will be challenging and require a non-trivial amount of work. Web scraping is a lengthy process that often requires the building of specific processes per website, and thorough cleaning of the data afterwards. **Our aim is to gather around 500 documents per persona**.

7.2 Linguistic, geographical and cultural constraints

To facilitate the resource gathering and to assure all members of our group can understand every source, we are forced, at least in the first stages of the project, to only work on english-speaking countries, especially the U.S.A.

This will inevitably set the prospective of the whole project: from an European point of view the American political spectrum is skewed towards the right. Labels like 'left wing' or 'right wing' take on multiple meanings and become not so reliable, and multiple issues are approached differently based on the country.

7.3 Time and resource constraints

Models used in this project will be open source and of limited size. Techniques like fine-tuning had to be discarded as the research team does not have access to adequate resources and the time investment would not be compatible with the project schedule

7.4 Future Developments

Currently the project focuses on In-Context Learning, achieved through fewshot prompting and iteratively refined prompt engineering, combined with RAG on a curated corpus.

To achieve better results with a constant amount of resources, we could implement quantization techniques to employ more powerful models, more sensible to fewshot approaches and more capable in general.

Furthermore, after reaching satisfying results with our current project structure, we could move on with a multi-agent architecture, eg. one model could handle the isolation of relevant information from the user query, another model could handle the RAG mechanic returning the top k chunks, a third model could further work on this chunks, etc. .

Additionally, working on the European context would be exciting and more impactful, as the amount of resources is limited compared to the American context. We would probably orient ourselves towards the countries of origin of this research group: Austria and Italy. If needed, the data would be machine translated to english (compromise quality for efficiency).

8 Appendix

We already did some research on tools and libraries that we could use for our project:

- **Vector Database:** `https://www.trychroma.com/`
- **Langchain (for agentic web search, text embeddings and RAG):** `https://docs.langchain.com/`
- **User Interface:** `https://docs.chainlit.io/get-started/overview`